

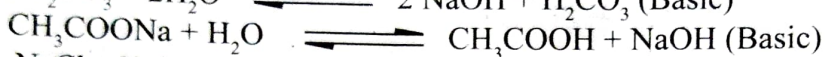
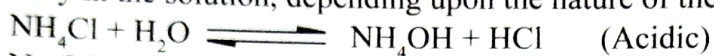
Experiment No. 16

Aim: To determine the pH and nature of salt solution (NH_4Cl , Na_2CO_3 , CH_3COONa , NaCl) using pH paper or Universal indicator

Apparatus: Five Test tubes, dropper, test tube stand, etc.

Chemicals: 1 gram of each salt NH_4Cl , Na_2CO_3 , CH_3COONa , NaCl and pH paper or universal indicator (with pH values and color strip).

Theory: Salts when dissolved in water, undergo hydrolysis and produce acidity, alkalinity (basicity) or neutrality in the solution, depending upon the nature of the salts.



Procedure:

1. Dissolve 1 gm of each salt in 10 mL of distilled water in four different clean test tubes marked A, B, C and D.
2. Add two to three drops of Universal indicator or dip one small strip of pH paper in each of the test tubes marked A, B, C and D.
3. Observe the color developed in the solution or pH paper strips and note the corresponding pH values. (Take one unknown salt and prepare its solution and find out its pH)

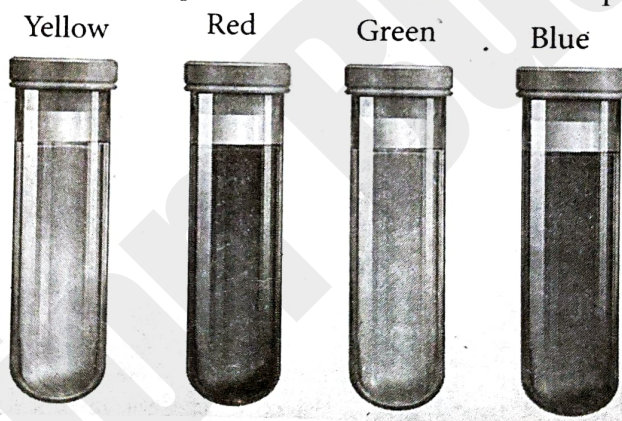


Fig. 16.1 Color of salts

Observation and result table:

Test Tube	Salt solution	Color of pH paper/color of solution	pH of solution	Nature of solution
A	NH_4Cl	Yellow	7	acid
B	Na_2CO_3	Dark green	8.5	Basic
C	CH_3COONa	Blue Green	4.5	Basic
D	NaCl	Light Green	6	neutral
E	Unknown			

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type of questions.

1. The pH of aqueous solution of NH_4Cl salt is
a. > 7 ☐ b. equal to 7 ☐ c. < 7 ☒ d. between 7 to 14 ☐
2. The nature of aqueous solution of CH_3COONa is
a. basic ☒ b. acidic ☐ c. neutral ☐ d. amphoteric ☐
3. The pH of aqueous solution of NaCl is
a. > 7 ☐ b. equal to 7 ☐ c. < 7 ☒ d. zero ☐
4. When pH paper is dipped in the aqueous solution of Na_2CO_3 the pH paper develop ----- color.
a. yellow ☐ b. green ☐ c. blue ☒ d. orange ☐
5. The nature of aqueous solution of Na_2CO_3
a. basic ☒ b. acidic ☐ c. neutral ☐ d. amphoteric ☐
6. A strip of pH paper is dipped in aqueous solution of NH_4Cl the colour develops on pH paper is -----
a. yellow ☒ b. green ☐ c. blue ☐ d. red ☐
7. The following salt does NOT undergo hydrolysis.
a. NH_4Cl ☐ b. HCl ☒ c. Na_2CO_3 ☐ d. CH_3COONa ☐

Short answer questions

1. Write chemical reaction for the hydrolysis of a salt of weak acid and strong base.

Ans. $\text{H}_2\text{CO}_3 + 2\text{NaOH} \rightleftharpoons \text{Na}_2\text{CO}_3 + 2\text{H}_2\text{O}$

2. Write name and chemical formula of a salt of strong acid and strong base.

Ans. Sodium chloride (NaCl)

3. When water is added in NaCl , it will undergo hydrolysis or not? Why?

Ans. NaCl is a salt of a strong acid HCl and strong base NaOH . It does not undergoes hydrolysis so there is no reaction of water between NaCl salt ion.

Remark and sign of teacher: